

Dynamic Guideline: WHO-Based Hypertension - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO hypertension clinical guideline: “Guideline for the pharmacological treatment of hypertension in adults”, published **24 August 2021**. A WHO summary version was published **14 June 2022**. No newer WHO replacement hypertension treatment guideline was identified as of **29 April 2026** [WHO publication page](#), [WHO summary publication page](#).

- **Target population:** Adults with confirmed hypertension who are **non-pregnant**, appropriately diagnosed, and counselled on lifestyle modification [WHO full PDF](#).
- **Scope:** Pharmacological treatment after confirmed diagnosis, including treatment-initiation thresholds, first-line drug classes, combination therapy, follow-up intervals, BP targets, and task sharing [WHO publication page](#).
- **Important limitation:** The WHO 2021 guideline **does not address BP measurement, diagnostic technique, or hypertensive crisis**; it applies after hypertension is confirmed, for example after elevated BP on two different days [WHO full PDF](#).
- **Global burden context:** WHO estimated that in **2024, 1.4 billion adults aged 30–79 years** had hypertension worldwide; about **600 million** were unaware, **630 million** were diagnosed and treated, and **320 million** had controlled hypertension [WHO hypertension fact sheet](#).

Exactly 5 WHO Hypertension-Related Guidelines or Technical Guidance Documents Synthesized

#	WHO document	Date	Role in hypertension management	Source
1	Guideline for the pharmacological treatment of hypertension in adults	24 August 2021	Core WHO adult hypertension pharmacological-treatment guideline	WHO publication page
2	HEARTS: Tool for the development of a consensus protocol for treatment of hypertension	1 April 2021	Supports standardized hypertension treatment protocols	WHO HEARTS consensus protocol tool
3	HEARTS: Risk-based CVD management	13 July 2020	Total cardiovascular-risk assessment and CVD management in primary care	WHO Risk-based CVD management module
4	HEARTS: Healthy-lifestyle counselling	2 May 2018	Lifestyle counselling within primary-care CVD prevention	WHO hypertension report page
5	HEARTS: Evidence-based treatment protocols	2 May 2018	Standardized evidence-based CVD/hypertension treatment protocols	WHO hypertension report page

1.2 Key Recommendations

GRADE interpretation used in this report:

- **High certainty:** Very confident that the true effect is close to the estimate.
- **Moderate certainty:** Moderately confident; true effect is likely close but may differ.
- **Low certainty:** Limited confidence; true effect may be substantially different.

- **Very low certainty:** Very limited confidence.
- **Strong recommendation:** Benefits clearly outweigh harms/burden for most patients.
- **Conditional recommendation:** Applies depending on patient values, feasibility, resources, comorbidities, or setting.

WHO recommendation or management point	GRADE Evidence Quality	Strength	Clinical application	Sources
Start pharmacological antihypertensive treatment in adults with confirmed hypertension and SBP ≥ 140 mmHg or DBP ≥ 90 mmHg.	Moderate to High	Strong	Core global threshold for adult, non-pregnant patients after confirmed diagnosis.	WHO full PDF , NCBI Bookshelf , PMC policy-implications article
Start pharmacological treatment in adults with existing cardiovascular disease and SBP 130–139 mmHg.	Moderate to High	Strong	Treat lower BP range when established CVD is present.	WHO full PDF , PMC policy-implications article
Consider pharmacological treatment in adults without CVD but with high cardiovascular risk, diabetes, or CKD and SBP 130–139 mmHg.	Low	Conditional	Individualize based on total risk, tolerability, access, and patient preference.	PMC policy-implications article
Use one of the following first-line drug classes: thiazide/thiazide-like diuretic, ACE inhibitor or ARB, or long-acting dihydropyridine calcium-channel blocker.	Moderate*	Strong/Conditional*	Choose based on comorbidity, contraindications, cost, availability, and local protocol.	WHO PDF excerpt , NCBI Bookshelf
Prefer initial combination therapy , ideally as a single-pill combination , for adults requiring pharmacological treatment.	Moderate	Conditional	Improves adherence and persistence; use WHO-preferred first-line classes.	NCBI Bookshelf , PMC policy-implications article
For adults with hypertension without comorbidities , treat to <140/90 mmHg.	Moderate	Strong	Minimum WHO global treatment target; consider lower individualized targets if tolerated and supported by newer evidence.	PMC policy-implications article
Follow up monthly after treatment initiation or medication change until target BP is reached.	Low†	Conditional†	Practical titration interval for primary-care programs.	Aristo Pharma PDF summary of WHO guideline

WHO recommendation or management point	GRADE Evidence Quality	Strength	Clinical application	Sources
Once BP is controlled, follow up every 3–6 months .	Low†	Conditional†	Stable treated patients can be monitored less frequently if adherence and control are maintained.	Aristo Pharma PDF summary of WHO guideline
Use standardized, drug- and dose-specific hypertension protocols in large-scale programs.	Moderate†	Strong†	Supports procurement, training, task sharing, quality reporting, and scale-up.	WHO HEARTS consensus protocol tool
Incorporate total cardiovascular-risk assessment and lifestyle counselling through HEARTS primary-care implementation.	Low to Moderate†	Strong/Conditional†	Especially relevant in resource-limited settings and population-level CVD prevention.	WHO Risk-based CVD management module, WHO hypertension fact sheet

* The retrieved summaries clearly identify the WHO-recommended drug classes but do not provide the exact official GRADE line for this item.

† GRADE rating is analyst-assessed from the retrieved evidence and implementation nature of the recommendation, because the exact official WHO GRADE wording was not available in the supplied summaries.

1.3 Guideline Development Methodology

- **Evidence review process:** WHO addressed prespecified clinical and public-health questions, including treatment thresholds, laboratory testing, CVD risk assessment, first-line drug classes, combination therapy, target BP, reassessment intervals, and treatment by nonphysician health workers [PMC policy-implications article](#).
- **Evidence synthesis:** Systematic reviews were used where available; primary research was examined when systematic reviews were not available [PMC policy-implications article](#).
- **GRADE methodology:** The WHO guideline used the **GRADE** approach to rate certainty of evidence and strength of recommendations [PMC policy-implications article](#).
- **Consensus method:** Recommendations reflect the guideline development group’s judgement about the balance of desirable and undesirable effects, feasibility, equity, acceptability, and resource implications.
- **Conflict of interest management:** The available summaries confirm that disclosure information exists in guideline summaries, but the detailed conflict-of-interest declarations were not extractable from the supplied results [Medscape summary adapted from WHO](#).

1.4 Required WHO Youth-Engagement Guideline-Process Item

The most recent WHO publication matching the requested title pattern is “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, published **11 December 2025** [WHO publication page](#). It is linked to the **WHO Abortion care guideline**, published **8 March 2022**, not to hypertension [WHO Abortion care guideline](#).

Item	Finding	GRADE Evidence Quality	Strength / Relevance	Source
Meaningful youth engagement evidence brief	Describes youth input through consultations and youth-led convening in the 2022 WHO Abortion Care Guideline process.	Low†	Conditional relevance to future hypertension guideline development, not current hypertension guidance.	WHO publication page , WHO webinar page
Applicability to hypertension	No supplied WHO source showed that this youth-engagement model was applied to a WHO hypertension guideline.	Very Low	No hypertension-specific recommendation change.	WHO publication page

† GRADE rating is analyst-assessed because this is process evidence rather than treatment-effect evidence.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Evidence window: After the WHO 2021 hypertension pharmacological-treatment guideline release on **24 August 2021** through **29 April 2026**.

Evidence sources reviewed from the supplied summaries included:

- WHO guideline and HEARTS materials.
- WHO hypertension fact sheet and global hypertension reports.
- ESC 2024, ESH 2023, and ACC/AHA 2025 guideline summaries and reviews.
- Systematic reviews and meta-analyses on intensive BP control, older adults, resistant hypertension, and pharmacist-led care.
- RCT evidence summaries including PATHWAY-2, CHAP, renal denervation sham-controlled trials, and intensive BP-control trials cited in guideline updates.
- Evidence and guideline commentary on home BP monitoring, potassium-based salt substitutes, and renal denervation.

Inclusion criteria:

1. Adult hypertension diagnosis, treatment, monitoring, or implementation evidence.
2. Randomized trials, systematic reviews, meta-analyses, high-quality guideline updates, or authoritative institutional guidance.
3. Evidence published or summarized after the WHO 2021 guideline release.

Exclusion criteria:

1. Non-human studies.
2. Evidence unrelated to BP diagnosis, treatment, control, adherence, or implementation.
3. Hypertension-in-pregnancy evidence was considered separately because the WHO 2021 adult pharmacological-treatment guideline excludes pregnant patients.

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality	Strength / implication	Sources
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Study / source	Design	Key findings	GRADE Quality	Strength / implication	Sources
Intensive BP control in older hypertensive adults	Systematic review and meta-analysis of RCTs	Intensive BP targets reduced adverse cardiovascular outcomes in older hypertensive adults without clear increase in all-cause mortality, renal failure, or serious adverse events.	Moderate	Conditional: consider lower targets in fit older adults if tolerated.	Systematic review/meta-analysis
JSH2025-commissioned meta-analysis in adults ≥75 years	Meta-analysis of 7 RCTs	Intensive SBP lowering to <130 mmHg reduced composite cardiovascular events, RR 0.61 , 95% CI 0.40–0.94 , and all-cause mortality, RR 0.72 , 95% CI 0.56–0.93 .	Moderate	Conditional to strong for independent older adults; individualize for frailty and orthostasis.	Nature guideline comparison
Intensive BP targets in high-risk and diabetes populations	RCT evidence cited in 2025 guideline implementation review, including 2024 Lancet and 2025 NEJM trials	Newer trials support more intensive BP lowering in selected high-risk patients, including diabetes, while emphasizing tolerability monitoring.	Moderate	Conditional: lower targets may be appropriate in high-risk patients if tolerated.	PMC guideline implementation article
ESC 2024 guideline	Contemporary guideline update	Defines hypertension as ≥140/90 mmHg , introduces “elevated BP” 120–139/70–89 mmHg , and targets SBP 120–129 mmHg for most treated adults if tolerated.	Moderate	Conditional: more intensive targets for selected adults; retain diagnostic threshold of 140/90.	ACC key points on ESC 2024, PMC review of ESC 2024
ESH 2023 guideline	Contemporary guideline update	Maintains hypertension diagnosis around >140/90 mmHg and uses age-stratified targets, including more lenient targets in older adults.	Moderate	Conditional: supports individualized targets by age, CKD, and frailty.	ACC review of ESH 2023

Study / source	Design	Key findings	GRADE Quality	Strength / implication	Sources
ACC/AHA 2025 guideline	Contemporary guideline update	Replaces 2017 guideline, incorporates evidence through early 2025, emphasizes accurate measurement, standardized protocols, and reaffirms target <130/80 mmHg .	Moderate	Conditional/strong depending population; more intensive than WHO minimum target.	AHA Professional Heart Daily, ACC 2025 clinical guidance article
PATHWAY-2 resistant hypertension trial	Double-blind crossover RCT, 335 patients	In patients on ACEi/ARB + CCB + thiazide-type diuretic, spironolactone was superior to doxazosin, bisoprolol, and placebo for BP control.	High	Strong: use spironolactone as preferred fourth-line therapy when potassium and renal function permit.	Cleveland Clinic Journal of Medicine
Chlorthalidone vs hydrochlorothiazide evidence	Small RCT plus meta-analyses	Chlorthalidone reduced ambulatory and nighttime BP more than hydrochlorothiazide; meta-analyses found greater BP lowering with chlorthalidone.	Moderate	Conditional: prefer thiazide-like diuretics when feasible, monitor electrolytes.	Cleveland Clinic Journal of Medicine
Apparent resistant hypertension adherence review	Systematic review, 9 studies, 747 patients	Partial nonadherence occurred in 13%–46% and complete nonadherence in 2%–35% of apparent resistant hypertension cases.	Moderate	Strong: assess adherence before diagnosing true resistant hypertension.	Cleveland Clinic Journal of Medicine

Study / source	Design	Key findings	GRADE Quality	Strength / implication	Sources
Home BP monitoring with cointerventions	Guideline evidence synthesis of RCT meta-analyses	Home BP monitoring alone modestly lowers BP; larger and more clinically meaningful reductions occur when paired with education, telehealth, or algorithmic medication titration.	Moderate	Strong: use home BP monitoring with structured cointerventions where feasible.	2025 AHA/ACC guideline
Potassium-based salt substitutes	Guideline recommendation based on trial evidence	Potassium-based salt substitutes can help prevent or treat elevated BP/hypertension, especially where household salt use is high; caution in CKD or potassium-retaining drugs.	Moderate	Conditional: use when hyperkalemia risk is low and monitoring is possible.	2025 AHA/ACC guideline
Renal denervation evidence	Sham-controlled RCTs, meta-analyses, consensus statements	Later sham-controlled trials show modest but consistent BP reductions; long-term clinical outcome evidence remains lacking.	Moderate for BP lowering; Low for clinical outcomes	Conditional: consider only in selected uncontrolled/resistant hypertension after optimized therapy and adherence assessment.	SCAI/NKF expert roundtable , ESC/EAPCI consensus PDF , NHS England evidence review PDF
CHAP trial in mild chronic hypertension during pregnancy	Multicenter RCT	Treating mild chronic hypertension in pregnancy reduced severe-feature superimposed preeclampsia and medically indicated preterm birth; pregnancy remains outside WHO 2021 adult non-pregnant scope.	High	Strong for pregnancy-specific care; not directly applicable to WHO adult non-pregnant guideline.	Medscape hypertension management review

Study / source	Design	Key findings	GRADE Quality	Strength / implication	Sources
Pharmacist-led hypertension management in hypertension plus diabetes	2025 systematic review/meta-analysis of RCTs	Pharmacist interventions improved hypertension management; estimates comparable to earlier diabetes-focused review showing SBP reduction around -6.2 mmHg and DBP reduction around -4.5 mmHg .	Moderate	Conditional: integrate pharmacists/team-based care where systems allow.	BMC Health Services Research
Non-pharmacologic interventions network meta-analysis protocol	BMJ Open protocol	Planned review of RCTs from 2014–2024; no pooled results yet.	Very Low	No recommendation change until results are available.	BMJ Open

2.3 Evidence Gaps and Future Research Needs

- 1. **WHO diagnostic guidance gap:** The WHO 2021 pharmacological-treatment guideline does not cover BP measurement or diagnosis. Newer ESC/ESH/ACC/AHA guidelines emphasize standardized office measurement and out-of-office confirmation, but WHO-specific global diagnostic guidance remains a need [WHO full PDF](#), [ACC review of ESH 2023](#).
- 2. **Optimal lower BP targets in diverse populations:** Evidence increasingly supports intensive BP control, but uncertainty remains for frail older adults, patients with orthostatic hypotension, multimorbidity, advanced CKD, and older adults with diabetes [Systematic review/meta-analysis](#).
- 3. **Long-term renal denervation outcomes:** BP reductions are increasingly supported, but hard cardiovascular outcome trials and durability data remain limited [SCAI/NKF expert roundtable](#).
- 4. **Implementation in low- and middle-income countries:** WHO reports that two-thirds of adults with hypertension live in low- and middle-income countries; scalable approaches for detection, treatment, medication access, adherence, and monitoring remain essential [WHO hypertension fact sheet](#).
- 5. **Youth engagement in hypertension guideline development:** WHO has published a youth-engagement case study for the Abortion Care Guideline, but no supplied source shows equivalent youth engagement in WHO hypertension guideline development [WHO publication page](#).
- 6. **Lifestyle and salt-substitution implementation:** Potassium-based salt substitutes are promising but require better implementation strategies and hyperkalemia-safety pathways, especially in CKD and patients using potassium-retaining drugs [2025 AHA/ACC guideline](#).

2.4 Updated Recommendations Based on New Evidence

Original WHO recommendation	New evidence impact	Updated recommendation as of 29 April 2026	GRADE Quality	Strength
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Original WHO recommendation	New evidence impact	Updated recommendation as of 29 April 2026	GRADE Quality	Strength
Start medication at confirmed SBP ≥ 140 or DBP ≥ 90 mmHg.	ESC 2024 and ESH 2023 continue to use approximately 140/90 mmHg as the hypertension threshold, while ACC/AHA uses a lower diagnostic framework.	Retain WHO global initiation threshold $\geq 140/90$ mmHg for broad implementation, while considering treatment at SBP 130–139 mmHg in patients with CVD, diabetes, CKD, or high risk.	Moderate to High	Strong
Treat adults with existing CVD and SBP 130–139 mmHg.	Newer intensive-target evidence supports lower BP treatment in high-risk populations if tolerated.	Continue WHO recommendation; consider individualized target closer to $<130/80$ mmHg or SBP 120–129 mmHg in selected high-risk adults if tolerated and monitored.	Moderate	Strong/Conditional
WHO target for uncomplicated hypertension: $<140/90$ mmHg.	ESC 2024, ACC/AHA 2025, and meta-analyses support lower targets in many patients, but tolerability and resources matter.	Use $<140/90$ mmHg as the minimum global control target; consider $<130/80$ mmHg or SBP 120–129 mmHg in selected patients at high CV risk, younger patients, and fit older adults, avoiding symptomatic hypotension.	Moderate	Conditional
Prefer first-line classes: thiazide/thiazide-like diuretic, ACEi/ARB, or long-acting dihydropyridine CCB.	Newer evidence supports thiazide-like potency, especially chlorthalidone, with monitoring.	Continue first-line class recommendation; consider thiazide-like diuretics such as chlorthalidone where available, with electrolyte and renal monitoring.	Moderate	Conditional
Use initial combination therapy, preferably single-pill combinations.	Newer ACC/AHA and implementation evidence continues to favor combinations and adherence-focused care.	Continue use of single-pill combinations when available and affordable, especially when baseline BP is substantially above target or rapid control is needed.	Moderate	Conditional
Follow monthly until BP target reached, then every 3–6 months.	Home BP monitoring plus education, telehealth, and algorithmic titration improves control.	Combine clinic follow-up with validated home BP monitoring and structured cointerventions when feasible.	Moderate	Strong

Original WHO recommendation	New evidence impact	Updated recommendation as of 29 April 2026	GRADE Quality	Strength
WHO HEARTS supports standardized protocols and primary-care scale-up.	WHO reports HEARTS implementation in about 30 countries and more than seven million people on treatment using simple algorithms.	Prioritize standardized national protocols, reliable medicine supply, team-based care, and monitoring dashboards.	Moderate	Strong
Resistant hypertension not specifically emphasized in WHO 2021 summary.	PATHWAY-2 supports spironolactone; adherence reviews show frequent nonadherence.	Before labeling resistant hypertension, confirm out-of-office BP, exclude white-coat effect, assess adherence, optimize triple therapy, then add spironolactone if renal function and potassium allow.	High for spironolactone BP effect; Moderate for adherence prevalence	Strong
Device therapy not part of WHO 2021 routine recommendations.	Renal denervation shows modest BP lowering in sham-controlled trials but lacks hard outcome evidence.	Do not use renal denervation routinely; consider only for selected uncontrolled/resistant hypertension after optimized medication, adherence assessment, and specialist shared decision-making.	Moderate for BP; Low for outcomes	Conditional
Lifestyle counselling through HEARTS.	Salt-substitute guidance has strengthened, but CKD/hyperkalemia risks remain.	Encourage sodium reduction, healthy diet, physical activity, weight management, tobacco cessation, and reduced harmful alcohol use; consider potassium-based salt substitutes only when hyperkalemia risk is low.	Moderate	Conditional
Pregnancy excluded from WHO 2021 adult guideline.	CHAP supports treating mild chronic hypertension in pregnancy.	Manage pregnancy separately under pregnancy-specific protocols; do not directly apply WHO adult non-pregnant targets.	High	Strong for pregnancy-specific care
Youth engagement not included in WHO hypertension guideline process.	WHO 2025 youth-engagement evidence brief shows feasibility in abortion-care guideline development.	Future WHO hypertension guideline updates should consider structured engagement of adolescents and young adults, especially for prevention, screening, adherence, and transition-to-adult-care issues.	Low	Conditional

Overall conclusion: As of **29 April 2026**, the WHO 2021 pharmacological-treatment guideline remains the latest WHO hypertension treatment guideline. Newer evidence supports retaining WHO's globally implementable **≥140/90 mmHg** treatment threshold while increasingly favoring **more intensive, individualized BP targets** in selected high-risk and fit

older adults; stronger use of **home BP monitoring with cointerventions**; careful assessment of **adherence** before diagnosing resistant hypertension; **spironolactone** as preferred fourth-line therapy for resistant hypertension; and cautious, selective consideration of **renal denervation** only after optimized medical management.